Supplementary material

Supplementary information

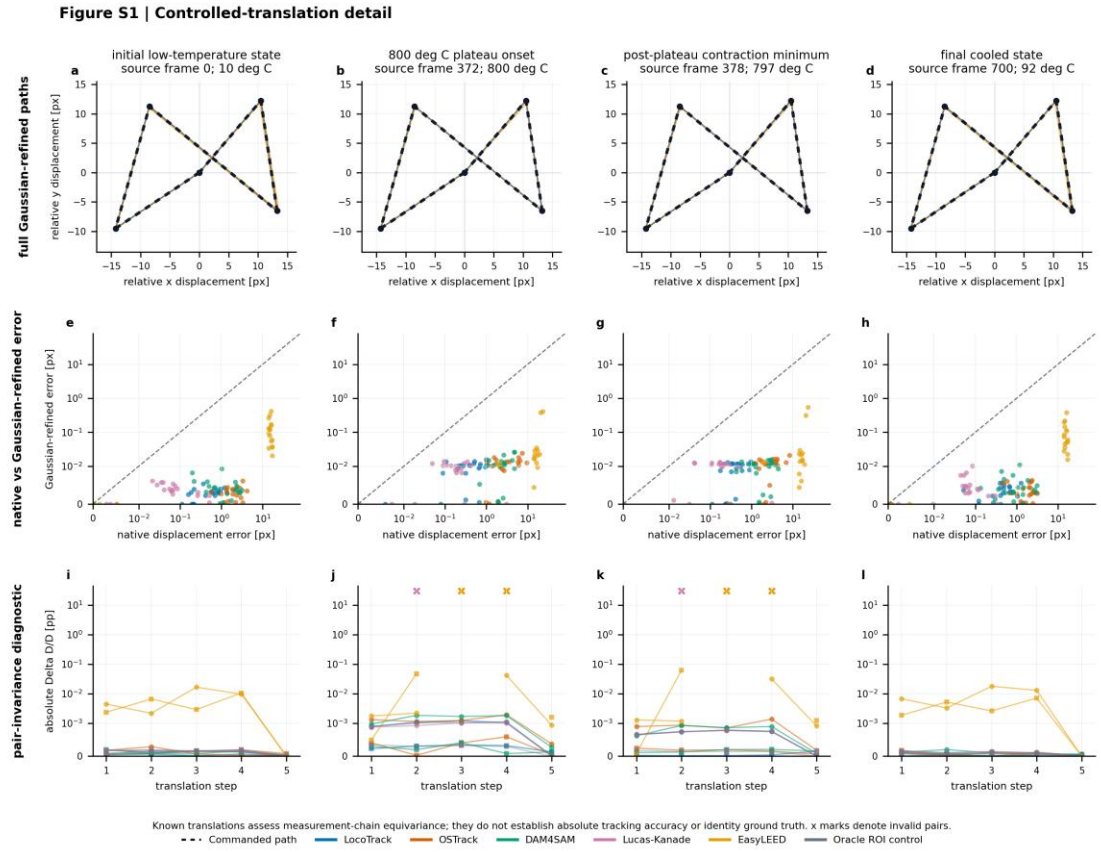
2026-08-02

Table of Contents

Figure S1 — Controlled-translation detail 2
Table S1 — Software, models, versions, hashes and environments 2
Native proposal availability and preprocessing contract 15
Table S2 — Gaussian gates, fixed opportunity sets and conditional denominators 16
Table S3 — Ag(111) outcomes by scan–pair and leave-one-case-out 28
Table S4 — Cu(111) pairwise physical endpoints 33
Table S5 — Controlled translations by source trial 35
Table S6 — Measurement-window and capture-contract sensitivity 40
Table S7 — Cross-regime numerical summary 52
Additional supplementary figures 58

This working supplement contains deterministic display items linked from the main manuscript. Internal archive identifiers and machine-specific paths are kept in committed provenance records rather than reader-facing tables.

Figure S1 — Controlled-translation detail



Complete controlled-translation trial paths and diagnostics

Figure S1. Controlled-translation detail. Columns are the four independently initialized source states from one Cu(111) series. The top row overlays the commanded path and every accepted Gaussian-refined spot path for all five trackers and the oracle ROI measurement control. The middle row retains every available native-versus-refined displacement-error observation. The bottom row shows both pair-invariance trajectories; crosses mark invalid pair measurements, whose fixed and conditional denominators are reported in Table S5. Known translations assess measurement-chain equivariance and common-estimator behavior, not absolute tracking accuracy or identity ground truth.

Table S1 — Software, models, versions, hashes and environments

Table S1. Software, models, versions, hashes and environments. Software and model identities and the run-frozen records in blocks A and B come from the completed execution contracts. Where a method-level configuration was not frozen during a run, block B instead reports a post-run normalized record derived from the archived executed

source; the status column distinguishes these cases. The package inventories in block C were captured on 2026-07-29 after the completed runs from the retained local environments; they document the preserved runtimes, but this record does not establish bitwise reproducibility of GPU inference. The retained execution host was `Microsoft Windows NT 10.0.26200.0, 64-bit` with an NVIDIA GeForce RTX 4070 Laptop GPU (8188 MiB, driver 610.62).

A. Software and model identity

Component	Study role/regime	Software version	Upstream revision	Configuration/variant	Frozen artifact	Artifact type	Artifact SHA-256	Environment
TrackLEED common measurement	All regimes and reported positions	Source files identified by SHA-256	not recorded separately	48 × 48 raw-image ROI; two-dimensional elliptical-Gaussian estimator	peak_analyzer.py; common-measurement adapter	implementation source	5d6af3da893300f4f8bb36196b73e7ad1584566d1f12f5db9a93bf2bb7013416; c4f41ad5debdd5a6e932345553e3127570d83da08045682b8a9b3940efd55364	netcat
LocoTrack	Main five-method matrix	not tagged	3385a233a270160b253a69050c6a3934527f7775	Base, offline mode	locotrack_base.ckpt	model checkpoint	8a5adbaeb610d1f06adfbc7c9076b66f727d674c0fd1d668890201cf3339736c	locotrack-pytorch
OSTrack	Main five-method matrix	not tagged	33b5e12586216b7fd0e95d255bd01ba44cbec759	ViT-B/256 MAE-CE, epoch 300	OSTrack_ep0300.pth.tar	model checkpoint	30c2147dd5799c7d78f5c114bf9ebaf00fa69638568865e6ade3d8cb04ea380f	ostrack

Component	Study role/regime	Software version	Upstream revision	Configuration/variant	Frozen artifact	Artifact type	Artifact SHA-256	Environment
DAM4SAM Large	Cu(111) and controlled translations	not tagged	9c954504b39ebca4c412f207be0787c26bfac85a	sam21pp-L / Hiera Large	sam2.1_hiera_large.pt	model checkpoint	2647878d5dfa5098f2f8649825738a9345572bae2d4350a2468587ece47dd318	dam4sam
Lucas–Kanade	Main five-method matrix	TrackLEED implementation	not applicable	Pyramidal skip-anchor; 41 × 41 window; four pyramid levels; forward–backward threshold 8 px	trackleed/analysis/tracking.py	implementation source	554bc16ec6c2ebc0493b3eac65ccd8180b0f52cc65e56702ba2d60ff12c1275e	netcat
EasyLEED	Main five-method matrix and restricted external reference	2.5.2	bbe4acd2b5eeb0f5d50a8d018780ace3afd7ff73	LEED motion model and Kalman filter; 12 px tracking radius in the method input grid	TrackLEED EasyLEED adapters	implementation source	be5701c5a3f86736ae096ad3321980756e532ef457af13059ebf916ebb07e288	easyleed

Component	Study role/regime	Software version	Upstream revision	Configuration/variant	Frozen artifact	Artifact type	Artifact SHA-256	Environment
SAM2.1	Controlled SAM2 family	2.1	2b90b9f5ceec907a1c18123530e92e794ad901a4	Hiera Base+ vanilla propagation	sam2.1_hiera_base_plus.pt	model checkpoint	a2345aede8715ab1d5d31b4a509fb160c5a4af1970f199d9054ccfb746c004c5	sam2
SAMURAI	Controlled SAM2 family	not tagged	76ba195984892b0d1e3db5d9c9f90bb62175680a	Hiera Base+ with motion-aware memory	sam2.1_hiera_base_plus.pt	model checkpoint	a2345aede8715ab1d5d31b4a509fb160c5a4af1970f199d9054ccfb746c004c5	samurai
DAM4SAM Base+	Independent energy scans and controlled SAM2 family	not tagged	9c954504b39ebca4c412f207be0787c26bfac85a	sam21pp-B / Hiera Base+	sam2.1_hiera_base_plus.pt	model checkpoint	a2345aede8715ab1d5d31b4a509fb160c5a4af1970f199d9054ccfb746c004c5	dam4sam
ViPErLEED	Restricted whole-pattern external reference	1.03	2ed1044b72bb43a936f9456cdb8b9846e0d23e96	Frozen 18-beam pattern and screen registration at 187.9 eV	ViPErLEED plugin JAR	runtime plugin	dd058af9ea79eaf6bea837efc6188b8f9b67dadf980b540d84826a33062c8a2d	viperleed-imagej

B. Configuration and implementation hashes

Component	Configuration-record SHA-256	Configuration-record status	Upstream configuration SHA-256	Implementation/adapter SHA-256
TrackLEED common measurement	not recorded separately	not recorded separately	not applicable	c4f41ad5debdd5a6e932345553e3127570d83da08045682b8a9b3940efd55364
LocoTrack	056238541ea2b81555857fd7746b4872e7f2dbfe9fe5418973bcef47b05446df; 111f6c16214e51ac3f0838e587c8a6f564c389594c874966c5bfa554717d7b97	run-frozen for Cu(111)/translations; post-run normalized from executed source for independent energy scans	b8cd20e5d5bc1bc5988ecde26ea182bc94de7a8d594bbf868e0f80defa9c86a7	23cf5087bb90e874e8de20f33f3bd48142a83bcadec8862c162fd11e5e0923e4; fc8e2aa3de8e1d7c24ad2d69c6bcb8c05151c9b8a540a1379134d1ba8bbb3eee
OSTrack	dc9bd17199ffc0aaea78fc00d0f68ba0b2b3047ef15fac30ec1c4997ed973736; d834f2be2cfb6f96bafee9fe2fef8b819e85bf69f8b6389d2b188e6d67d8fa4f	run-frozen for Cu(111)/translations; post-run normalized from executed source for independent energy scans	7eff86b7bef5082378d3daadf54decaa9d00cf95b02687c6b3bf1a79b56da701	33755330d3328801a42a807c35442302cb1438fad0167a39ab609dbafc52919c; fc8e2aa3de8e1d7c24ad2d69c6bcb8c05151c9b8a540a1379134d1ba8bbb3eee
DAM4SAM Large	d2ffbdc023b7c5be3238a2dca229ebd0bcb980d994b11ca85a2b124d1c85daa1	run-frozen executed configuration	742e07598bcab0b09012b3fdf0ed15fec2928c750916c7b226571e9deb3ec563	ee20f7898d6e74a6f9fc953c806b8e52cab1ba651772049b88849ea8c18c8586

Component	Configuration-record SHA-256	Configuration-record status	Upstream configuration SHA-256	Implementation/adapter SHA-256
Lucas–Kanade	e755aca300764ee823973948744fcdd1802a8a26d0009b00ef49911b6229e99e; bb4bbb2a166371124e37be32f43fd67dc3785dc5035ecd830ec8df50450a83d3	run-frozen for Cu(111)/translations; post-run normalized from executed source for independent energy scans	not applicable	554bc16ec6c2ebc0493b3eac65ccd8180b0f52cc65e56702ba2d60ff12c1275e; fc8e2aa3de8e1d7c24ad2d69c6bcb8c05151c9b8a540a1379134d1ba8bbb3eee
EasyLEED	54049d3d51c53f33ac8581366a4a979e4a25c2718d69f2b94e60970d6f99aecc; cbb16fba737a2c4fadea7643dc3cbfa5908603382adfee116002149b875a5494; 1ffc0c5e429095e78ac8d7d086e733661cc303fa6bdcec290c10b322e694c8ca	run-frozen for Cu(111)/translations; post-run normalized from executed sources and outputs for energy scans/external reference	not applicable	be5701c5a3f86736ae096ad3321980756e532ef457af13059ebf916ebb07e288; d11818a491ed44aa045622d06f6ccc5b4f2d1acb577e02b76b1d758b473ac31c; aea4de426b7c307d9230c3994ee88cc59579a95ae8bf0ce8079690e485f90933; 29ceb1e22e89a177a68c2af3f14ef6f331e2656daa83e71611dd4e9edcfbc23b

Component	Configuration-record SHA-256	Configuration-record status	Upstream configuration SHA-256	Implementation/adapter SHA-256
SAM2.1	123c6defcfc619877b0969ce23595998cfd6b95cac2e70e9dd5369560be78c46	post-run normalized from executed source	not recorded separately	2f6b6442e3aafafa913f78400dab15d4c7c715d93bb7577009dc75beb7b31797; c95eb72343da954378e083233b943e79beed78328e3695dc71b4a5f51aaa9547
SAMURAI	79d70dbb54fbc200bdace10d9b7117ded66e88219572e7791d28bf9486274cac	post-run normalized from executed source	not recorded separately	bf2933f3f3fc5ef31c0fc0e530d5f270469be83a8858552f04d2989c32f8ae21; c95eb72343da954378e083233b943e79beed78328e3695dc71b4a5f51aaa9547
DAM4SAM Base+	62e9316adac13b4b5fa72ec8b656668590d48bc150e387734373715b04b2b301	post-run normalized from executed source	not recorded separately	ee20f7898d6e74a6f9fc953c806b8e52cab1ba651772049b88849ea8c18c8586; c95eb72343da954378e083233b943e79beed78328e3695dc71b4a5f51aaa9547; fc8e2aa3de8e1d7c24ad2d69c6bcb8c05151c9b8a540a1379134d1ba8bbb3eee

Component	Configuration-record SHA-256	Configuration-record status	Upstream configuration SHA-256	Implementation/adapter SHA-256
ViPErLEED	not recorded separately	not recorded separately	not applicable	not applicable

C. Retained execution environments

Environment	Used for	Runtime	Core package versions	Execution mode/runtime	Frozen snapshot SHA-256
netcat	TrackLEED common Gaussian measurement, Lucas–Kanade, and result assembly	Python 3.12.9	NumPy 2.2.5; SciPy 1.15.2; OpenCV 4.11.0.86; scikit-image 0.25.2	CPU	14f756ccfe8c2f2781aa1ad8ea405d7c58151529ff476ae58b88efe0629e5668
locotrack-pytorch	LocoTrack inference	Python 3.11.15	PyTorch 2.11.0+cu128; torchvision 0.26.0+cu128; NumPy 2.4.4; OpenCV 4.13.0.92	CUDA; PyTorch build CUDA 12.8	812c5c9af7cd60058cd6b3bcdde8aac4dd90c4bc46568444f64c9f722d4b5bd5
ostrack	OSTrack inference	Python 3.8.20	PyTorch 1.13.1+cu117; torchvision 0.14.1+cu117; NumPy 1.24.4; OpenCV 4.13.0.92; timm 0.5.4	CUDA; PyTorch build CUDA 11.7	a4b9193c9cf4829f0f697ba35a2112d58011c4056b924a5976022226c3882b11
dam4sam	DAM4SAM Large and Base+ inference	Python 3.10.20	PyTorch 2.1.0+cu121; torchvision 0.16.0+cu121; NumPy 1.26.4; OpenCV 4.11.0.86; Hydra 1.3.2	CUDA; PyTorch build CUDA 12.1	1b14a78e06eff162d756fbbae9753f84371814d53639abc4275165aeb012147e

Environment	Used for	Runtime	Core package versions	Execution mode/runtime	Frozen snapshot SHA-256
sam2	Vanilla SAM2.1 controlled-family inference	Python 3.10.20	PyTorch 2.5.1+cu121; torchvision 0.20.1+cu121; NumPy 2.2.6; OpenCV 4.13.0.92; Hydra 1.3.2	CUDA; PyTorch build CUDA 12.1	22d4b6254ba42b3391c3df2c468b1fe8142cb86b6ffab67a0ce0026f0afc5bef
samurai	SAMURAI controlled-family inference	Python 3.10.20	PyTorch 2.5.1+cu121; torchvision 0.20.1+cu121; NumPy 1.26.4; OpenCV 4.11.0.86; Hydra 1.3.2	CUDA; PyTorch build CUDA 12.1	c2d2c6b2156ba94dd08b7a8cb68e99e6342d655b07431ebfd639010347a5d11a
easyleed	EasyLEED Kalman tracking	Python 3.11.15	NumPy 1.26.4; SciPy 1.17.1; scikit-image 0.26.0; PyQt5 5.15.11	CPU	e2f4a155c8696dd250f41b9cccb287c76be9fe88da9b3e77aee5dab3577f1539
viperleed-imagej	ViPErLEED whole-pattern tracking	ImageJ 1.54u; OpenJDK 1.8.0_345; ViPErLEED ImageJ plugin 1.03	not applicable	JVM/CPU	e64540325636048157605336981c63f3680b31b1c5808a1e1d49803f893d6ac3

Native proposal availability and preprocessing contract

This reader-facing contract complements the identity and hash inventory in Table S1. Records labelled as post-run normalized in Table S1 were transcribed from the archived executed source and outputs; that status is not upgraded to a run-frozen method configuration.

- **LocoTrack:** inference used the Base offline model on a 256 × 256 input. For the independent energy scans, native availability combined the returned occlusion logit o and expected-distance logit d as $P_{\mathrm{vis}} = [1 - \mathrm{sigmoid}(o)][1 - \mathrm{sigmoid}(d)]$, with $P_{\mathrm{vis}} > 0.5$ required. In the Cu(111) and translation regimes every finite point was available and this probability remained diagnostic only.
- **OSTrack:** inference used the frozen 256-pixel configuration. A finite box with positive width and height was available; its center was the proposal. The native box score was retained but did not gate or weight the common fit.
- **DAM4SAM and the controlled SAM2 family:** a finite nonempty mask was available and its pixel centroid was the proposal. The controlled SAM2, SAMURAI, and DAM4SAM paths used byte-identical checkpoint weights, the same one-object box prompt, the same bidirectional image order, and no later prompt or reset. Table S1 identifies the distinct DAM4SAM configurations used across regimes.
- **Lucas–Kanade:** the pyramidal skip-anchor implementation used a 41 × 41 window, maximum pyramid level 4, at most 50 iterations, convergence tolerance 0.01, and minimum eigenvalue 10^{-5}. Native acceptance required residual ≤ 30 and forward–backward closure error ≤ 8 px; normalized cross-correlation and centroid recentering were disabled. For the energy scans, each image was normalized to grayscale, high-pass filtered by subtracting a Gaussian blur with $\sigma = 5$, processed by a white top-hat with a disk of radius 5, renormalized, and converted to 8-bit input. A rejected image remained missing and the last successful image and point stayed as the later anchor without interpolation.
- **EasyLEED:** the native energy-scan contract used a 12-pixel tracking radius, input precision 2 px, native Gaussian R^2 threshold 0.8, validation parameter $\gamma = 8$, and window scaling. The exported finite Kalman coordinate was the proposal; native intensity and internal Gaussian output remained diagnostics and did not replace the common measurement.
- **ViPErLEED:** the restricted compatibility check used the frozen whole-pattern initialization recorded in Table S1. Its corresponding beam coordinates were proposal inputs to the common estimator, not external position ground truth.

Table S2 — Gaussian gates, fixed opportunity sets and conditional denominators

Table S2. Gaussian gates, fixed opportunity sets and conditional denominators. Block A lists the run-frozen acceptance contract and separates Gaussian acceptance from the subsequent energy-branch consistency classification. Block B retains every predefined opportunity in its fixed denominator, including unavailable proposals and rejected fits. Block C gives the method-dependent n used only for statistics conditional on available native outputs, accepted Gaussian measurements, or jointly branch-consistent pairs. Counts pool repeated observations of propagated trajectories; images, spot opportunities, pair images, and gaps are not statistically independent replicates.

A. Gaussian measurement and evaluation gates

Regime	Stage	Frozen criterion	Boundary and role	Failure handling
All experiments	Proposal-to-ROI conversion	Map the proposal to the original 1000 × 1000 image; round to the nearest pixel; place a 48 × 48 raw-intensity ROI.	The ROI must lie completely inside the image; it is rejected rather than clipped or padded.	No proposal or an out-of-bounds ROI produces a missing measurement.
Cu(111) and controlled translations	Numerical fit validity	Fit result, R^2, fitted parameters, and center uncertainties must be finite; center uncertainties must be nonnegative.	Gaussian-acceptance gate applied identically to all compared methods in this regime.	Failure produces a missing measurement.
Cu(111) and controlled translations	Fit quality	$R^2 \geq 0.85$ and amplitude $A \geq 0.05$.	Both thresholds are inclusive.	Failure produces a missing measurement.
Cu(111) and controlled translations	Fitted-center and width geometry	$-2 < x_c, y_c < 50$ px and $0.5 < \sigma_x, \sigma_y < 48$ px.	All four bounds are strict; coordinates are local to the 48 × 48 ROI.	Failure produces a missing measurement.
Energy scans	Fit initialization and solver bounds	Initialize the center at the smoothed intensity maximum within 16.32 px of the ROI center; bound each fitted center coordinate to ±16.32 px from the ROI center.	Solver constraint, not a post-fit performance metric.	Fit failure produces a missing measurement.
Energy scans	Fit quality and widths	$R^2 \geq 0.85$; amplitude $A \geq 0.012$; $0.6 \leq \sigma_x, \sigma_y \leq 13$ px.	All threshold bounds are inclusive.	Failure produces a missing measurement.

Regime	Stage	Frozen criterion	Boundary and role	Failure handling
Energy scans	Proposal-to-fit consistency	Euclidean distance between the proposal and Gaussian-refined center ≤ 15 px.	Gaussian-acceptance gate applied identically to all compared methods in this regime.	Failure produces a missing measurement.
Energy scans	Branch consistency	Euclidean distance between an accepted Gaussian center and the frozen expected radial branch ≤ 18 px.	Evaluation gate applied after Gaussian acceptance; it does not move the proposal or fitted center.	A fit beyond 18 px remains an accepted Gaussian fit but is classified as geometry-inconsistent.
All experiments	Missing-value policy	No interpolation, carry-forward, manual correction, or within-run reinitialization.	Applies after unavailable proposals, out-of-bounds ROIs, fit failures, or gate failures.	The predefined opportunity remains in its fixed denominator.
Cu(111) primary pairs	Relative-endpoint reference	D_ref is the median of valid pair distances in the first 10 images, requiring at least 8 valid distances.	Eligibility rule for each method–pair relative endpoint, not a Gaussian-acceptance gate.	If fewer than 8 of 10 distances are valid, the entire relative endpoint is unavailable.

B. Fixed opportunity denominators

Regime	Metric	Denominator class	Denominator per method	Unit	Construction	Replication/stratification	Scope note
Controlled translations	Spot proposal and Gaussian-measurement coverage	Fixed opportunity	80	spot–step opportunities	4 source-state trials × 4 spots × 5 non-seed steps	Four separately initialized trials drawn from one Cu(111) sequence	Failures remain in 80.
Controlled translations	Pair coverage and pair-invariance endpoint	Fixed opportunity	40	pair–step opportunities	4 source-state trials × 2 pairs × 5 non-seed steps	Four separately initialized trials drawn from one Cu(111) sequence	Failures remain in 40.
Cu(111) positive control	Spot proposal and Gaussian-measurement coverage	Fixed opportunity	2,804	spot–image opportunities	701 images × 4 frozen spots	One sequence; four spots	Failures remain in 2,804.
Cu(111) positive control	Primary relative pair endpoint	Fixed opportunity	701 per pair	pair–image opportunities	701 images for each of pairs 1–6 and 3–5	One sequence; two frozen pairs	The two pair strata are not independent series.

Regime	Metric	Denominator class	Denominator per method	Unit	Construction	Replication/stratification	Scope note
Cu(111) positive control	D_ref eligibility window	Fixed eligibility window	10 per method–pair	initial pair–image opportunities	First 10 images; at least 8 valid pair distances required	Applied separately to every method–pair endpoint	This gate determines whether the relative endpoint exists.
Cu(111) pair 2–4 diagnostic	Consecutive amplitude-only fit failures	Separate diagnostic	24	rejected Gaussian candidates	24 consecutive proposal-present candidates; all passed R^2, center, and width gates but failed amplitude	Earlier saved YOLO proposal trajectory; outside the five-method matrix	Diagnostic count, not a denominator for comparing the five methods.
Independent energy scans	Branch-consistent measurement coverage	Fixed opportunity	221	baseline-visible spot opportunities	Frozen tracker-independent baseline across four scan–pair cases	Two independently acquired series; four frozen scan–pair cases	Every method uses the same 221 opportunities.

Regime	Metric	Denominator class	Denominator per method	Unit	Construction	Replication/stratification	Scope note
Independent energy scans	Post-gap branch-consistent recovery	Fixed opportunity	24	evaluable natural I(V) gaps	Baseline gaps bracketed by visible measurements on the same branch	Two independently acquired series; four frozen scan–pair cases	Every method uses the same 24 gaps.
Controlled SAM2 family	Branch-consistent measurement coverage	Fixed opportunity	182	baseline-visible spot opportunities	Frozen baseline for two pairs in one memory-analysis scan	One series; two frozen pairs	Mechanism experiment, not a population ranking.
Controlled SAM2 family	Post-gap branch-consistent recovery	Fixed opportunity	13	evaluable natural I(V) gaps	Baseline gaps bracketed by visible measurements on the same branch	One series; two frozen pairs	Mechanism experiment, not a population ranking.
Restricted external reference	Branch-image coverage	Fixed opportunity	60	branch–image opportunities	15 images × 4 frozen branches	Subset of the memory-analysis scan; two pairs	60 opportunities for EasyLEED and 60 for ViPErLEED.

Regime	Metric	Denominator class	Denominator per method	Unit	Construction	Replication/stratification	Scope note
Restricted external reference	Predefined branch-gap checks	Fixed opportunity	4	branch-gap checks	One check for each frozen branch	Subset of the memory-analysis scan; two pairs	Compatibility check, not external position ground truth.
Cu(111) frozen visual audit	Combined overlay review	Audit sample	24	combined overlays	12 temporal quantiles + 12 temperature-cycle anchors selected independently of method output	One Cu(111) sequence	Audit only; not a metric denominator and not used to correct outcomes.

C. Method-dependent conditional denominators

Regime	Metric	Denominator class	Unit	Method-specific or paired n	Conditioning basis	Conditioning rule
Controlled translations	Native-proposal displacement error	Conditional available proposals	available native spot–step proposals	LocoTrack=80; OSTrack=80; DAM4SAM=80; Lucas–Kanade=76; EasyLEED=80	Only opportunities with an available native proposal	Reported only alongside fixed proposal-coverage denominator 80.
Controlled translations	Gaussian-refined displacement error	Conditional accepted measurements	accepted spot–step measurements	LocoTrack=80; OSTrack=80; DAM4SAM=80; Lucas–Kanade=76; EasyLEED=74; Oracle ROI=80	Only opportunities with an accepted Gaussian measurement	Reported only alongside fixed measurement-coverage denominator 80.
Controlled translations	Pair-invariance error	Conditional jointly valid pairs	jointly valid pair–step measurements	LocoTrack=40; OSTrack=40; DAM4SAM=40; Lucas–Kanade=36; EasyLEED=34; Oracle ROI=40	Only pair opportunities with both Gaussian measurements accepted	Reported only alongside fixed pair denominator 40.

Regime	Metric	Denominator class	Unit	Method-specific or paired n	Conditioning basis	Conditioning rule
Independent energy scans	Geometry-inconsistent continuation	Conditional accepted fits	accepted Gaussian fits in all visibility regimes	LocoTrack=272; OSTrack=269; DAM4SAM=272 ; Lucas–Kanade=206; EasyLEED=222	Accepted Gaussian fits more than 18 px from the expected branch divided by all accepted fits	This denominator is not the 221 baseline-visible coverage denominator.
Independent energy scans	Absolute pair-distance residual	Conditional branch-consistent pairs	jointly valid and branch-consistent pair images	LocoTrack=91; OSTrack=90; DAM4SAM=91; Lucas–Kanade=68; EasyLEED=84	Both branches must pass proposal, Gaussian, and 18 px branch-consistency gates	Residual n is reported with the conditional statistic.
Controlled SAM2 family	Geometry-inconsistent continuation	Conditional accepted fits	accepted Gaussian fits in all visibility regimes	SAM2=149; SAMURAI=236; DAM4SAM=233	Accepted Gaussian fits more than 18 px from the expected branch divided by all accepted fits	This denominator is not the 182 baseline-visible coverage denominator.

Regime	Metric	Denominator class	Unit	Method-specific or paired n	Conditioning basis	Conditioning rule
Controlled SAM2 family	Pair-stratified absolute pair-distance residual	Conditional branch-consistent pairs	jointly valid and branch-consistent pair images	SAM2: pair 1=10, pair 2=9; SAMURAI: pair 1=38, pair 2=34; DAM4SAM: pair 1=36, pair 2=34	Both branches must pass proposal, Gaussian, and 18 px branch-consistency gates	Each pair-specific residual statistic is reported with its own n.
Restricted external reference	Native cross-method coordinate agreement	Conditional paired native outputs	paired branch–image correspondences	EasyLEED–ViPErLEED=60 paired correspondences	Both methods provide the predefined corresponding native branch coordinate	Native-output comparison precedes Gaussian refinement.
Restricted external reference	Common-Gaussian center agreement	Conditional jointly accepted fits	jointly accepted common centers	EasyLEED–ViPErLEED=38 jointly accepted centers	Both native outputs yield an accepted common Gaussian center for the same branch image	Reported alongside fixed branch-image denominator 60 per method.

Table S3 — Ag(111) outcomes by scan–pair and leave-one-case-out

Table S3. Ag(111) outcomes by scan–pair and leave-one-case-out. Coverage is branch-consistent measurements / baseline-visible opportunities; post-gap branch-consistent recovery is recovered / evaluable natural I(V) gaps; geometry inconsistency is accepted Gaussian fits more than 18 px from the expected branch / all accepted Gaussian fits across all visibility regimes. The conditional residual is the median absolute pair-distance residual and is evaluated only when both branches are valid and branch-consistent, with its surviving pair-frame n shown separately. The four scan–pair cases are descriptive strata from two independently acquired scans, not four independent acquisitions.

A. Outcomes by frozen scan–pair case

Scan	Pair	Method	Branch-consistent coverage	Post-gap branch-consistent recovery	Geometry-inconsistent accepted fits	Conditional median absolute pair-distance residual [px]	Conditional n
Scan A	pair 1	LocoTrack	44/44	5/5	9/57	1.983	18
Scan A	pair 1	OSTrack	44/44	5/5	9/58	1.822	19
Scan A	pair 1	DAM4SAM	44/44	5/5	9/58	1.825	19
Scan A	pair 1	Lucas-Kanade	43/44	5/5	5/49	1.983	16
Scan A	pair 1	EasyLEED	42/44	5/5	0/42	1.526	15
Scan A	pair 2	LocoTrack	48/50	7/7	7/57	1.561	19
Scan A	pair 2	OSTrack	49/50	6/7	7/58	1.647	20
Scan A	pair 2	DAM4SAM	49/50	7/7	7/58	1.650	20
Scan A	pair 2	Lucas-Kanade	40/50	4/7	0/40	1.560	14
Scan A	pair 2	EasyLEED	48/50	5/7	0/48	1.650	18
Scan B	pair 1	LocoTrack	58/60	5/6	10/78	1.336	26
Scan B	pair 1	OSTrack	57/60	5/6	10/77	1.166	25
Scan B	pair 1	DAM4SAM	57/60	5/6	10/77	1.164	25
Scan B	pair 1	Lucas-Kanade	47/60	4/6	9/59	1.131	15
Scan B	pair 1	EasyLEED	57/60	6/6	0/58	1.131	23
Scan B	pair 2	LocoTrack	67/67	6/6	10/80	2.068	28
Scan B	pair 2	OSTrack	66/67	5/6	8/76	1.990	26
Scan B	pair 2	DAM4SAM	66/67	6/6	10/79	2.035	27

Scan	Pair	Method	Branch-consistent coverage	Post-gap branch-consistent recovery	Geometry-inconsistent accepted fits	Conditional median absolute pair-distance residual [px]	Conditional n
Scan B	pair 2	Lucas-Kanade	58/67	4/6	0/58	2.034	23
Scan B	pair 2	EasyLEED	67/67	6/6	4/74	2.065	28

B. Leave-one-case-out sensitivity

Each of the four frozen scan–pair cases is omitted once. The remaining three cases are pooled and only the count-based coverage, post-gap branch-consistent recovery, and geometry-inconsistency rates are recomputed. The displayed extrema across the four omissions are a sensitivity check, not a confidence interval. Conditional case-median residuals are not reaggregated.

Method	Coverage range	Post-gap branch-consistent recovery range	Geometry-inconsistency range
LocoTrack	97.4–98.8%	94.1–100.0%	12.6–13.5%
OSTrack	97.2–98.8%	84.2–88.9%	11.8–13.5%
DAM4SAM	97.2–98.8%	94.1–100.0%	12.6–13.6%
Lucas-Kanade	81.9–87.6%	63.2–76.5%	3.4–9.5%
EasyLEED	95.5–97.5%	88.9–100.0%	0.0–2.4%

Table S4 — Cu(111) pairwise physical endpoints

Table S4. Cu(111) pairwise physical endpoints. Values are ΔR/R = ΔD/D in percent after the native proposal from each method entered the common Gaussian measurement. Every method produced 701/701 valid endpoints for each primary pair. The 800 °C plateau is the median of a predefined five-image interval. The post-plateau minimum was selected as the lowest value after that interval; it occurred in the common image acquired at 797 °C and is therefore outcome-selected. The final cooled state is the predefined final image acquired at 92 °C. D_ref is the median valid pair distance in the first ten images, with at least eight required. Cross-method concordance after common refinement is not external absolute accuracy or an overall tracker ranking. The two pairs come from one series, are not independent replicates, and do not establish crystallographic anisotropy.

Pair	Method	Valid endpoints	800 °C plateau median ΔR/R [%]	Post-plateau minimum (797 °C) ΔR/R [%]	Final cooled state (92 °C) ΔR/R [%]
1–6	LocoTrack	701/701	−0.686595	−0.726452	0.121606
1–6	OSTrack	701/701	−0.686676	−0.726482	0.121625
1–6	DAM4SAM	701/701	−0.686244	−0.726195	0.121691
1–6	Lucas–Kanade	701/701	−0.686551	−0.726440	0.121609
1–6	EasyLEED	701/701	−0.686375	−0.726389	0.121681
3–5	LocoTrack	701/701	−1.570535	−1.579948	−0.152413
3–5	OSTrack	701/701	−1.570833	−1.579795	−0.152417
3–5	DAM4SAM	701/701	−1.570300	−1.579962	−0.152478
3–5	Lucas–Kanade	701/701	−1.570362	−1.579880	−0.152338
3–5	EasyLEED	701/701	−1.570755	−1.580123	−0.152343

Table S5 — Controlled translations by source trial

Table S5. Controlled translations by source trial. Each deterministic six-image trial contains one seed and five evaluation steps on the same closed path, providing 20 spot-step and 10 pair-step fixed opportunities per method. Displacement summaries are conditional on accepted Gaussian measurements; pair invariance is conditional on jointly valid pairs, and both conditional n values are shown. The four source trials are descriptive strata drawn from one Cu(111) series, not four independent acquisitions. The oracle ROI is a measurement control, not a tracker. Known translations assess measurement-chain equivariance and common-estimator behavior, not absolute tracking accuracy and not identity ground truth.

Trial	T [deg C]	Method	Proposal / constructed-ROI coverage	Gaussian measurement coverage	Conditional median / P95 / max displacement error [px] (n)	Pair coverage	Max absolute Delta D/D [pp] (n)
initial low-temperature state	10	LocoTrack	20/20	20/20	0.0031 / 0.0044 / 0.0054 (n=20)	10/10	0.000181 (n=10)
initial low-temperature state	10	OSTrack	20/20	20/20	0.0031 / 0.0046 / 0.0061 (n=20)	10/10	0.000290 (n=10)
initial low-temperature state	10	DAM4SAM	20/20	20/20	0.0037 / 0.0084 / 0.0094 (n=20)	10/10	0.000201 (n=10)
initial low-temperature state	10	Lucas-Kanade	20/20	20/20	0.0036 / 0.0058 / 0.0062 (n=20)	10/10	0.000188 (n=10)
initial low-temperature state	10	EasyLEED	20/20	20/20	0.0695 / 0.3250 / 0.4187 (n=20)	10/10	0.016820 (n=10)
initial low-temperature state	10	Oracle ROI control	20/20	20/20	0.0036 / 0.0058 / 0.0062 (n=20)	10/10	0.000188 (n=10)
800 deg C plateau onset	800	LocoTrack	20/20	20/20	0.0106 / 0.0141 / 0.0159 (n=20)	10/10	0.001229 (n=10)
800 deg C plateau onset	800	OSTrack	20/20	20/20	0.0121 / 0.0186 / 0.0244 (n=20)	10/10	0.001952 (n=10)

Trial	T [deg C]	Method	Proposal / constructed-ROI coverage	Gaussian measurement coverage	Conditional median / P95 / max displacement error [px] (n)	Pair coverage	Max absolute Delta D/D [pp] (n)
800 deg C plateau onset	800	DAM4SAM	20/20	20/20	0.0110 / 0.0261 / 0.0261 (n=20)	10/10	0.001869 (n=10)
800 deg C plateau onset	800	Lucas-Kanade	18/20	18/20	0.0099 / 0.0136 / 0.0140 (n=18)	8/10	0.001042 (n=8)
800 deg C plateau onset	800	EasyLEED	20/20	17/20	0.0176 / 0.3880 / 0.4066 (n=17)	7/10	0.047021 (n=7)
800 deg C plateau onset	800	Oracle ROI control	20/20	20/20	0.0104 / 0.0137 / 0.0140 (n=20)	10/10	0.001147 (n=10)
post-plateau contraction minimum	797	LocoTrack	20/20	20/20	0.0107 / 0.0130 / 0.0136 (n=20)	10/10	0.000794 (n=10)
post-plateau contraction minimum	797	OSTrack	20/20	20/20	0.0125 / 0.0170 / 0.0211 (n=20)	10/10	0.001397 (n=10)

Trial	T [deg C]	Method	Proposal / constructed-ROI coverage	Gaussian measurement coverage	Conditional median / P95 / max displacement error [px] (n)	Pair coverage	Max absolute Delta D/D [pp] (n)
post-plateau contraction minimum	797	DAM4SAM	20/20	20/20	0.0109 / 0.0154 / 0.0159 (n=20)	10/10	0.000934 (n=10)
post-plateau contraction minimum	797	Lucas-Kanade	18/20	18/20	0.0114 / 0.0129 / 0.0134 (n=18)	8/10	0.000784 (n=8)
post-plateau contraction minimum	797	EasyLEED	20/20	17/20	0.0139 / 0.3598 / 0.5412 (n=17)	7/10	0.062943 (n=7)
post-plateau contraction minimum	797	Oracle ROI control	20/20	20/20	0.0106 / 0.0128 / 0.0134 (n=20)	10/10	0.000784 (n=10)
final cooled state	92	LocoTrack	20/20	20/20	0.0033 / 0.0056 / 0.0057 (n=20)	10/10	0.000130 (n=10)

Trial	T [deg C]	Method	Proposal / constructed-ROI coverage	Gaussian measurement coverage	Conditional median / P95 / max displacement error [px] (n)	Pair coverage	Max absolute Delta D/D [pp] (n)
final cooled state	92	OSTrack	20/20	20/20	0.0031 / 0.0060 / 0.0063 (n=20)	10/10	0.000182 (n=10)
final cooled state	92	DAM4SAM	20/20	20/20	0.0039 / 0.0087 / 0.0109 (n=20)	10/10	0.000201 (n=10)
final cooled state	92	Lucas-Kanade	20/20	20/20	0.0039 / 0.0084 / 0.0102 (n=20)	10/10	0.000161 (n=10)
final cooled state	92	EasyLEED	20/20	20/20	0.0520 / 0.2251 / 0.3810 (n=20)	10/10	0.017988 (n=10)
final cooled state	92	Oracle ROI control	20/20	20/20	0.0039 / 0.0084 / 0.0102 (n=20)	10/10	0.000161 (n=10)

Table S6 — Measurement-window and capture-contract sensitivity

Table S6. Measurement-window and capture-contract sensitivity. For the independent Ag(111) stress test, coverage uses the fixed denominator 221 and post-gap branch-consistent recovery uses the fixed denominator 24. Geometry inconsistency is conditional on all accepted fits, while the pair-distance residual retains its method- and variant-specific conditional n. Reader-facing residuals are rounded to three decimals; full-precision values remain in the accompanying CSV and XLSX exports. No comparative rank or automatic configuration selection is implied. The asterisk marks only the frozen central contract.

Variant codebook.

- `R32-P12`: ROI 32 px; proposal-to-fit 12 px.
- `R32-P15`: ROI 32 px; proposal-to-fit 15 px.
- `R32-P18`: ROI 32 px; proposal-to-fit 18 px.
- `R48-P12`: ROI 48 px; proposal-to-fit 12 px.
- `R48-P15*`: frozen central contract.
- `R48-P18`: ROI 48 px; proposal-to-fit 18 px.
- `R64-P12`: ROI 64 px; proposal-to-fit 12 px.
- `R64-P15`: ROI 64 px; proposal-to-fit 15 px.
- `R64-P18`: ROI 64 px; proposal-to-fit 18 px.
- `B12`: branch tolerance 12 px.
- `B24`: branch tolerance 24 px.
- `R2-80`: R^2 minimum 0.80.
- `R2-90`: R^2 minimum 0.90.
- `A0.75`: amplitude gate multiplier 0.75.
- `A1.25`: amplitude gate multiplier 1.25.

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
R32-P12	EasyLEED	R32, P12; B18, R2-85, A1.00	209/221	21/24	2/223	1.627 / 3.659 (n=93)
R32-P12	LocoTrack	R32, P12; B18, R2-85, A1.00	220/221	24/24	38/286	1.776 / 5.759 (n=103)
R32-P12	OSTrack	R32, P12; B18, R2-85, A1.00	212/221	21/24	36/275	1.691 / 5.591 (n=95)
R32-P12	DAM4SAM	R32, P12; B18, R2-85, A1.00	217/221	24/24	38/285	1.755 / 5.784 (n=102)
R32-P12	Lucas–Kanade	R32, P12; B18, R2-85, A1.00	190/221	17/24	14/215	1.678 / 3.990 (n=77)
R32-P15	EasyLEED	R32, P15; B18, R2-85, A1.00	212/221	21/24	4/234	1.627 / 3.659 (n=93)
R32-P15	LocoTrack	R32, P15; B18, R2-85, A1.00	220/221	24/24	38/286	1.776 / 5.759 (n=103)
R32-P15	OSTrack	R32, P15; B18, R2-85, A1.00	215/221	21/24	36/279	1.711 / 5.567 (n=99)

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
R32-P15	DAM4SAM	R32, P15; B18, R2-85, A1.00	217/221	24/24	38/285	1.755 / 5.784 (n=102)
R32-P15	Lucas–Kanade	R32, P15; B18, R2-85, A1.00	190/221	17/24	14/215	1.678 / 3.990 (n=77)
R32-P18	EasyLEED	R32, P18; B18, R2-85, A1.00	212/221	21/24	4/239	1.627 / 3.659 (n=93)
R32-P18	LocoTrack	R32, P18; B18, R2-85, A1.00	220/221	24/24	38/286	1.776 / 5.759 (n=103)
R32-P18	OSTrack	R32, P18; B18, R2-85, A1.00	216/221	22/24	36/280	1.722 / 5.561 (n=100)
R32-P18	DAM4SAM	R32, P18; B18, R2-85, A1.00	217/221	24/24	38/285	1.755 / 5.784 (n=102)
R32-P18	Lucas–Kanade	R32, P18; B18, R2-85, A1.00	190/221	17/24	14/215	1.678 / 3.990 (n=77)
R48-P12	EasyLEED	R48, P12; B18, R2-85, A1.00	206/221	21/24	2/209	1.650 / 3.662 (n=82)

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
R48-P12	LocoTrack	R48, P12; B18, R2-85, A1.00	217/221	23/24	36/272	1.837 / 5.973 (n=91)
R48-P12	OSTrack	R48, P12; B18, R2-85, A1.00	211/221	20/24	34/262	1.737 / 5.650 (n=84)
R48-P12	DAM4SAM	R48, P12; B18, R2-85, A1.00	216/221	23/24	36/272	1.788 / 5.973 (n=91)
R48-P12	Lucas–Kanade	R48, P12; B18, R2-85, A1.00	188/221	17/24	14/206	1.764 / 4.442 (n=68)
R48-P15*	EasyLEED	central: R48, P15, B18, R2-85, A1.00	214/221	22/24	4/222	1.737 / 3.719 (n=84)
R48-P15*	LocoTrack	central: R48, P15, B18, R2-85, A1.00	217/221	23/24	36/272	1.837 / 5.973 (n=91)
R48-P15*	OSTrack	central: R48, P15, B18, R2-85, A1.00	216/221	21/24	34/269	1.770 / 5.937 (n=90)
R48-P15*	DAM4SAM	central: R48, P15, B18, R2-85, A1.00	216/221	23/24	36/272	1.788 / 5.973 (n=91)

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
R48-P15*	Lucas–Kanade	central: R48, P15, B18, R2-85, A1.00	188/221	17/24	14/206	1.764 / 4.442 (n=68)
R48-P18	EasyLEED	R48, P18; B18, R2-85, A1.00	219/221	22/24	6/237	1.837 / 5.595 (n=90)
R48-P18	LocoTrack	R48, P18; B18, R2-85, A1.00	217/221	23/24	36/272	1.837 / 5.973 (n=91)
R48-P18	OSTrack	R48, P18; B18, R2-85, A1.00	218/221	22/24	35/273	1.822 / 5.937 (n=93)
R48-P18	DAM4SAM	R48, P18; B18, R2-85, A1.00	216/221	23/24	36/272	1.788 / 5.973 (n=91)
R48-P18	Lucas–Kanade	R48, P18; B18, R2-85, A1.00	192/221	18/24	14/212	1.835 / 5.569 (n=73)
R64-P12	EasyLEED	R64, P12; B18, R2-85, A1.00	184/221	20/24	2/187	1.550 / 3.582 (n=65)
R64-P12	LocoTrack	R64, P12; B18, R2-85, A1.00	199/221	23/24	36/254	1.883 / 6.300 (n=77)

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
R64-P12	OSTrack	R64, P12; B18, R2-85, A1.00	191/221	20/24	34/241	1.725 / 6.009 (n=67)
R64-P12	DAM4SAM	R64, P12; B18, R2-85, A1.00	197/221	23/24	36/252	1.777 / 6.371 (n=75)
R64-P12	Lucas–Kanade	R64, P12; B18, R2-85, A1.00	170/221	17/24	14/188	1.774 / 5.171 (n=54)
R64-P15	EasyLEED	R64, P15; B18, R2-85, A1.00	192/221	21/24	4/200	1.634 / 3.685 (n=67)
R64-P15	LocoTrack	R64, P15; B18, R2-85, A1.00	199/221	23/24	36/254	1.883 / 6.300 (n=77)
R64-P15	OSTrack	R64, P15; B18, R2-85, A1.00	196/221	21/24	34/248	1.755 / 6.437 (n=73)
R64-P15	DAM4SAM	R64, P15; B18, R2-85, A1.00	197/221	23/24	36/252	1.777 / 6.371 (n=75)
R64-P15	Lucas–Kanade	R64, P15; B18, R2-85, A1.00	170/221	17/24	14/188	1.774 / 5.171 (n=54)

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
R64-P18	EasyLEED	R64, P18; B18, R2-85, A1.00	197/221	21/24	5/213	1.778 / 5.727 (n=73)
R64-P18	LocoTrack	R64, P18; B18, R2-85, A1.00	199/221	23/24	36/254	1.883 / 6.300 (n=77)
R64-P18	OSTrack	R64, P18; B18, R2-85, A1.00	198/221	22/24	35/252	1.814 / 6.329 (n=76)
R64-P18	DAM4SAM	R64, P18; B18, R2-85, A1.00	197/221	23/24	36/252	1.777 / 6.371 (n=75)
R64-P18	Lucas–Kanade	R64, P18; B18, R2-85, A1.00	172/221	18/24	14/192	1.882 / 5.968 (n=57)
B12	EasyLEED	R48, P15; B12, R2-85, A1.00	206/221	20/24	16/222	1.650 / 3.662 (n=82)
B12	LocoTrack	R48, P15; B12, R2-85, A1.00	203/221	20/24	69/272	1.550 / 3.568 (n=79)
B12	OSTrack	R48, P15; B12, R2-85, A1.00	204/221	20/24	64/269	1.527 / 3.542 (n=81)

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
B12	DAM4SAM	R48, P15; B12, R2-85, A1.00	202/221	20/24	69/272	1.499 / 3.548 (n=79)
B12	Lucas–Kanade	R48, P15; B12, R2-85, A1.00	183/221	17/24	23/206	1.736 / 3.555 (n=66)
B24	EasyLEED	R48, P15; B24, R2-85, A1.00	214/221	22/24	1/222	1.789 / 5.433 (n=87)
B24	LocoTrack	R48, P15; B24, R2-85, A1.00	217/221	23/24	19/272	1.939 / 7.670 (n=99)
B24	OSTrack	R48, P15; B24, R2-85, A1.00	216/221	21/24	19/269	1.829 / 7.034 (n=96)
B24	DAM4SAM	R48, P15; B24, R2-85, A1.00	216/221	23/24	19/272	1.873 / 7.670 (n=99)
B24	Lucas–Kanade	R48, P15; B24, R2-85, A1.00	188/221	17/24	9/206	1.764 / 4.442 (n=68)
R2-80	EasyLEED	R48, P15; B18, R2-80, A1.00	214/221	22/24	4/230	1.650 / 3.694 (n=92)

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
R2-80	LocoTrack	R48, P15; B18, R2-80, A1.00	221/221	24/24	37/286	1.834 / 5.779 (n=104)
R2-80	OSTrack	R48, P15; B18, R2-80, A1.00	217/221	22/24	34/278	1.743 / 5.725 (n=99)
R2-80	DAM4SAM	R48, P15; B18, R2-80, A1.00	217/221	24/24	37/281	1.788 / 5.832 (n=99)
R2-80	Lucas–Kanade	R48, P15; B18, R2-80, A1.00	190/221	17/24	14/215	1.733 / 3.901 (n=77)
R2-90	EasyLEED	R48, P15; B18, R2-90, A1.00	194/221	21/24	4/202	1.569 / 3.656 (n=69)
R2-90	LocoTrack	R48, P15; B18, R2-90, A1.00	200/221	23/24	36/255	1.837 / 6.221 (n=79)
R2-90	OSTrack	R48, P15; B18, R2-90, A1.00	195/221	21/24	34/247	1.743 / 6.437 (n=73)
R2-90	DAM4SAM	R48, P15; B18, R2-90, A1.00	196/221	23/24	36/251	1.788 / 6.365 (n=75)

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
R2-90	Lucas–Kanade	R48, P15; B18, R2-90, A1.00	170/221	17/24	14/188	1.788 / 5.107 (n=55)
A0.75	EasyLEED	R48, P15; B18, R2-85, A0.75 (threshold 0.009)	214/221	22/24	4/222	1.737 / 3.719 (n=84)
A0.75	LocoTrack	R48, P15; B18, R2-85, A0.75 (threshold 0.009)	217/221	23/24	36/272	1.837 / 5.973 (n=91)
A0.75	OSTrack	R48, P15; B18, R2-85, A0.75 (threshold 0.009)	216/221	21/24	34/269	1.770 / 5.937 (n=90)
A0.75	DAM4SAM	R48, P15; B18, R2-85, A0.75 (threshold 0.009)	216/221	23/24	36/272	1.788 / 5.973 (n=91)

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
A0.75	Lucas–Kanade	R48, P15; B18, R2-85, A0.75 (threshold 0.009)	188/221	17/24	14/206	1.764 / 4.442 (n=68)
A1.25	EasyLEED	R48, P15; B18, R2-85, A1.25 (threshold 0.015)	214/221	22/24	4/222	1.737 / 3.719 (n=84)
A1.25	LocoTrack	R48, P15; B18, R2-85, A1.25 (threshold 0.015)	217/221	23/24	36/272	1.837 / 5.973 (n=91)
A1.25	OSTrack	R48, P15; B18, R2-85, A1.25 (threshold 0.015)	216/221	21/24	34/269	1.770 / 5.937 (n=90)
A1.25	DAM4SAM	R48, P15; B18, R2-85, A1.25 (threshold 0.015)	216/221	23/24	36/272	1.788 / 5.973 (n=91)

Variant	Method	Contract change	Coverage	Recovery	Geometry inconsistency	Conditional median / P95 [px] (n)
A1.25	Lucas–Kanade	R48, P15; B18, R2-85, A1.25 (threshold 0.015)	188/221	17/24	14/206	1.764 / 4.442 (n=68)

Table S7 — Cross-regime numerical summary

Table S7. Cross-regime numerical summary. Each block answers a distinct validation question and retains its original denominators. The translation block uses 80 displaced-spot and 40 displaced-pair evaluation opportunities per method; the four initialization states reported in Table 1 are not displacement evaluations. Translation errors are conditional on accepted Gaussian measurements, with coverage shown first. The energy-scan residual is conditional on jointly branch-consistent pair frames and is shown with its n. Cu endpoint deviations are agreement with the framewise five-method median, not absolute accuracy. DAM4SAM used sam21pp-L for Cu(111) and controlled translations, and sam21pp-B for the energy scans. No composite score or rank is calculated.

Controlled translation

Method	Proposal coverage	Measurement coverage	Median / P95 displacement error [px]	Pair coverage	Maximum absolute Delta D/D [pp]
LocoTrack	80/80 (100.00%)	80/80 (100.00%)	0.0043 / 0.0130	40/40 (100.00%)	0.001229
OSTrack	80/80 (100.00%)	80/80 (100.00%)	0.0048 / 0.0168	40/40 (100.00%)	0.001952
DAM4SAM	80/80 (100.00%)	80/80 (100.00%)	0.0060 / 0.0154	40/40 (100.00%)	0.001869
Lucas–Kanade	76/80 (95.00%)	76/80 (95.00%)	0.0050 / 0.0127	36/40 (90.00%)	0.001042
EasyLEED	80/80 (100.00%)	74/80 (92.50%)	0.0265 / 0.3818	34/40 (85.00%)	0.062943

Cu(111) positive control

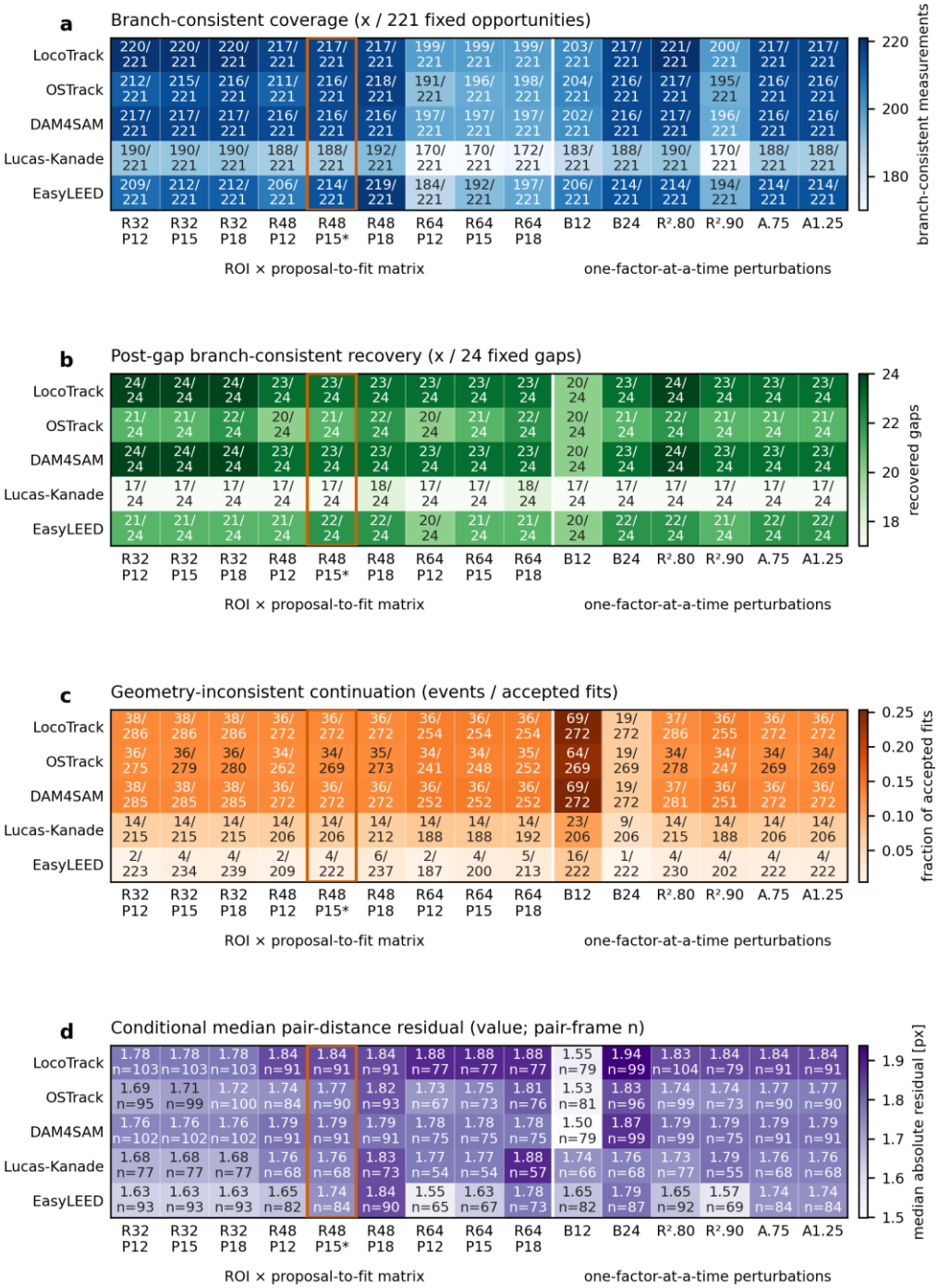
Method	Gaussian measurement coverage	Valid endpoints (pair 1-6; pair 3-5)	Median / maximum endpoint deviation [pp]
LocoTrack	2804/2804 (100.00%)	701/701; 701/701	0.000004 / 0.000299
OSTrack	2804/2804 (100.00%)	701/701; 701/701	0.000134 / 0.001752
DAM4SAM	2804/2804 (100.00%)	701/701; 701/701	0.000242 / 0.001564
Lucas–Kanade	2804/2804 (100.00%)	701/701; 701/701	0.000037 / 0.000462
EasyLEED	2804/2804 (100.00%)	701/701; 701/701	0.000256 / 0.000949

Independent Ag(111) energy-scan stress test

Method	Branch-consistent measurement coverage	Post-gap branch-consistent recovery	Geometry-inconsistent continuation	Conditional median pair-distance residual [px] (n)
LocoTrack	217/221 (98.19%)	23/24 (95.83%)	36/272 (13.24%)	1.837 (n=91)
OSTrack	216/221 (97.74%)	21/24 (87.50%)	34/269 (12.64%)	1.770 (n=90)
DAM4SAM	216/221 (97.74%)	23/24 (95.83%)	36/272 (13.24%)	1.788 (n=91)
Lucas–Kanade	188/221 (85.07%)	17/24 (70.83%)	14/206 (6.80%)	1.764 (n=68)
EasyLEED	214/221 (96.83%)	22/24 (91.67%)	4/222 (1.80%)	1.737 (n=84)

Additional supplementary figures

Figure S2a–d. Measurement-window and capture-contract sensitivity

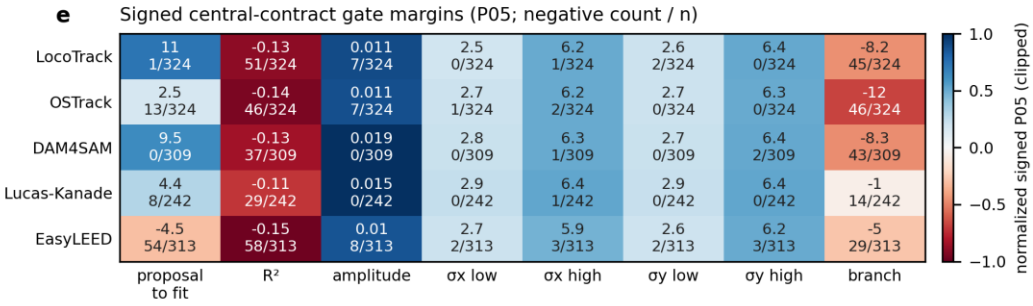


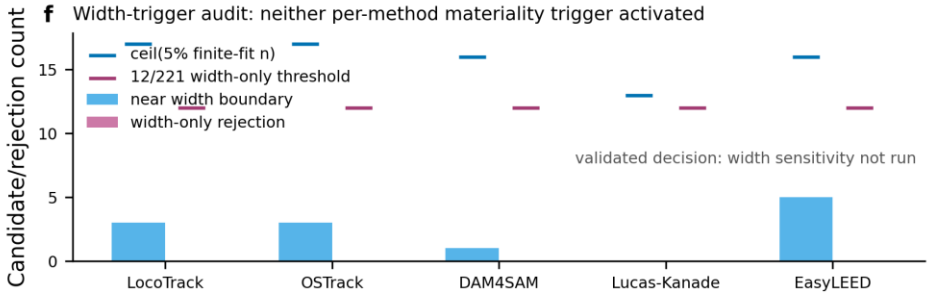
Codebook: R = ROI [px]; P = proposal-to-fit gate [px]; B = branch tolerance [px];
R² = fit threshold; A = amplitude multiplier; * = frozen central contract.

Figure S2. Panels a–d show the nine-element ROI × proposal-to-fit matrix and six one-factor-at-a-time perturbations for coverage, recovery, geometry inconsistency, and conditional pair-distance residual. Panel e shows the signed fifth-percentile central-contract margin for every acceptance gate, with negative counts and finite-margin n; its color uses the frozen gate-specific normalizer and is clipped only for display. Panel f shows the predefined per-method width-trigger thresholds and the validated decision not to run the conditional width sweep. The orange outline and asterisk mark the frozen central contract (48 px ROI and 15 px proposal-to-fit gate). The figure is descriptive and does not select a new central configuration.

Figure S2 remains one numbered display item with one caption. Its a–d and

Figure S2 (continued). Panels e–f





Codebook: R = ROI [px]; P = proposal-to-fit gate [px]; B = branch tolerance [px];
R² = fit threshold; A = amplitude multiplier; * = frozen central contract.

Figure S2 (continued).

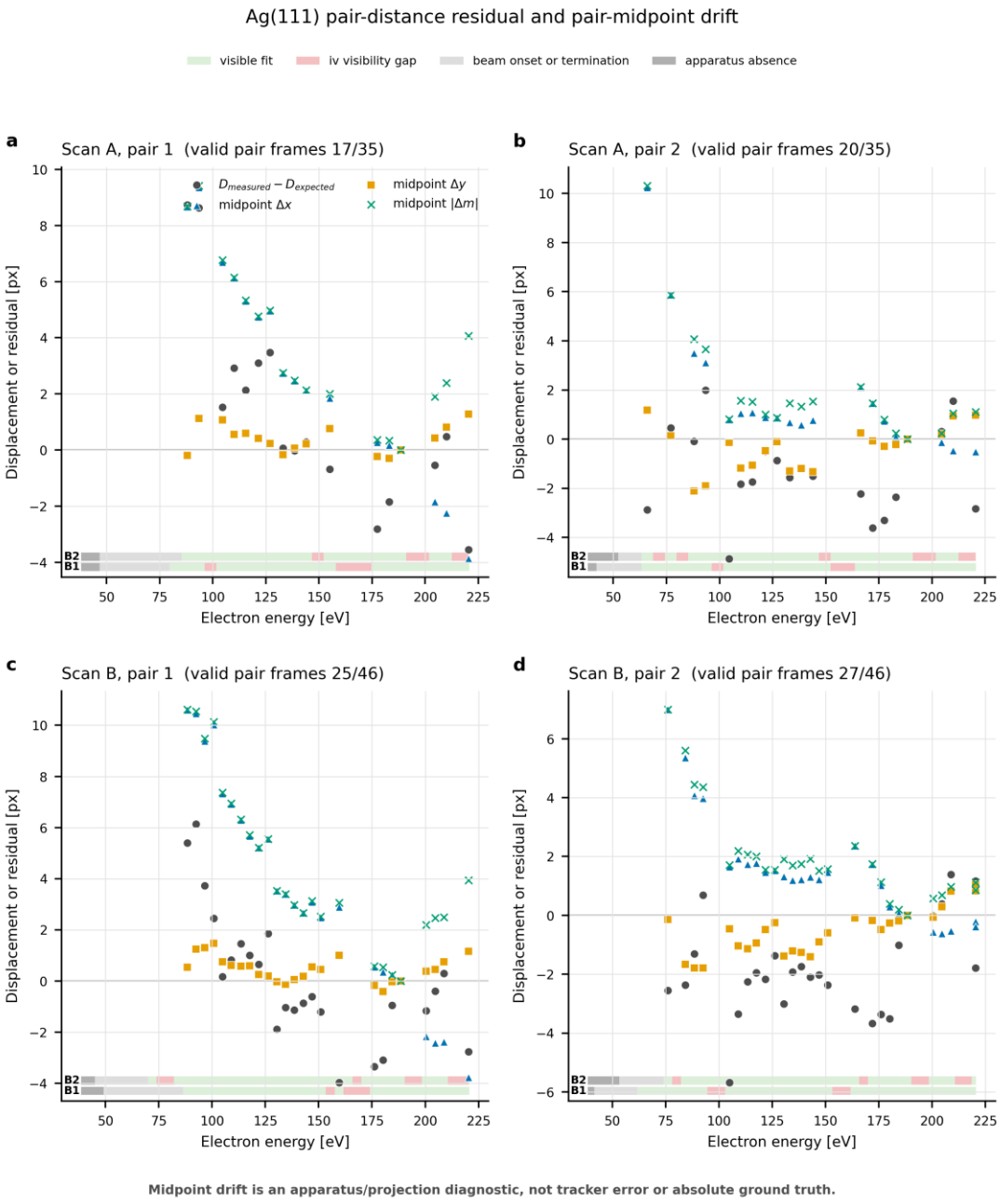


Figure S3. a–d, Pair-distance residual relative to the frozen radial law and pair-midpoint displacement for Scan A pair 1, Scan A pair 2, Scan B pair 1 and Scan B pair 2, respectively. Colored B1/B2 ribbons retain apparatus absence, beam onset/termination, baseline-visible and true I(V)-gap regimes. Midpoint motion is an apparatus/projection nuisance diagnostic kept separate from the translation-invariant pair-distance outcome; it is not absorbed into tracker error and does not turn the radial law into absolute position ground truth.

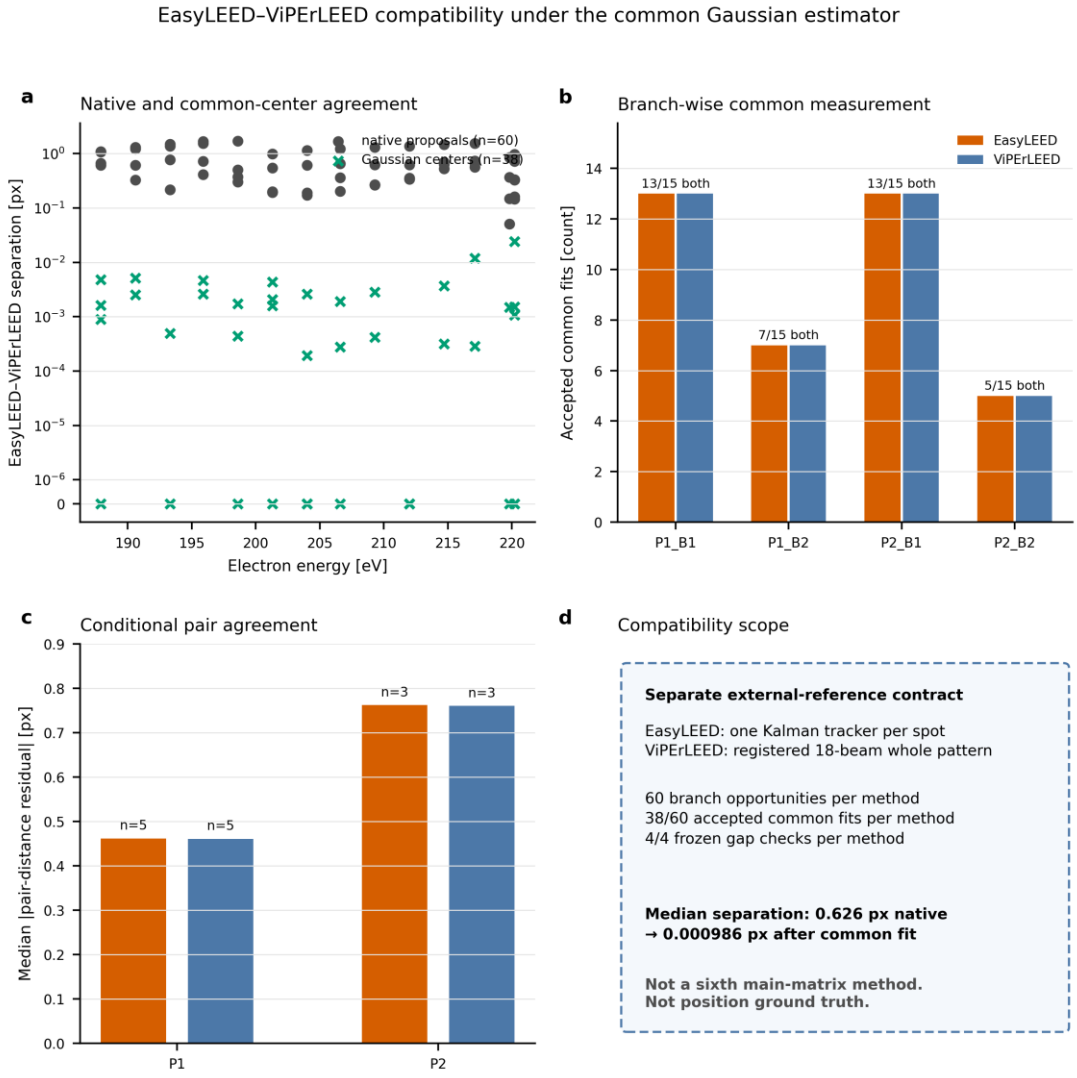


Figure S4. a, EasyLEED–ViPErLEED separation before and after common Gaussian refinement for matched branch/frame records. **b,** Accepted common fits for the four branches, each over 15 images. **c,** Conditional median absolute pair-distance residual with n=5 for P1 and n=3 for P2. **d,** Initialization, denominator and interpretation contract. ViPErLEED is not a sixth method in the main five-strategy matrix, and neither method supplies position or identity ground truth.

The source package is a byte-identical freeze of the completed high-energy compatibility artifact plus a normalized contract and hash provenance.

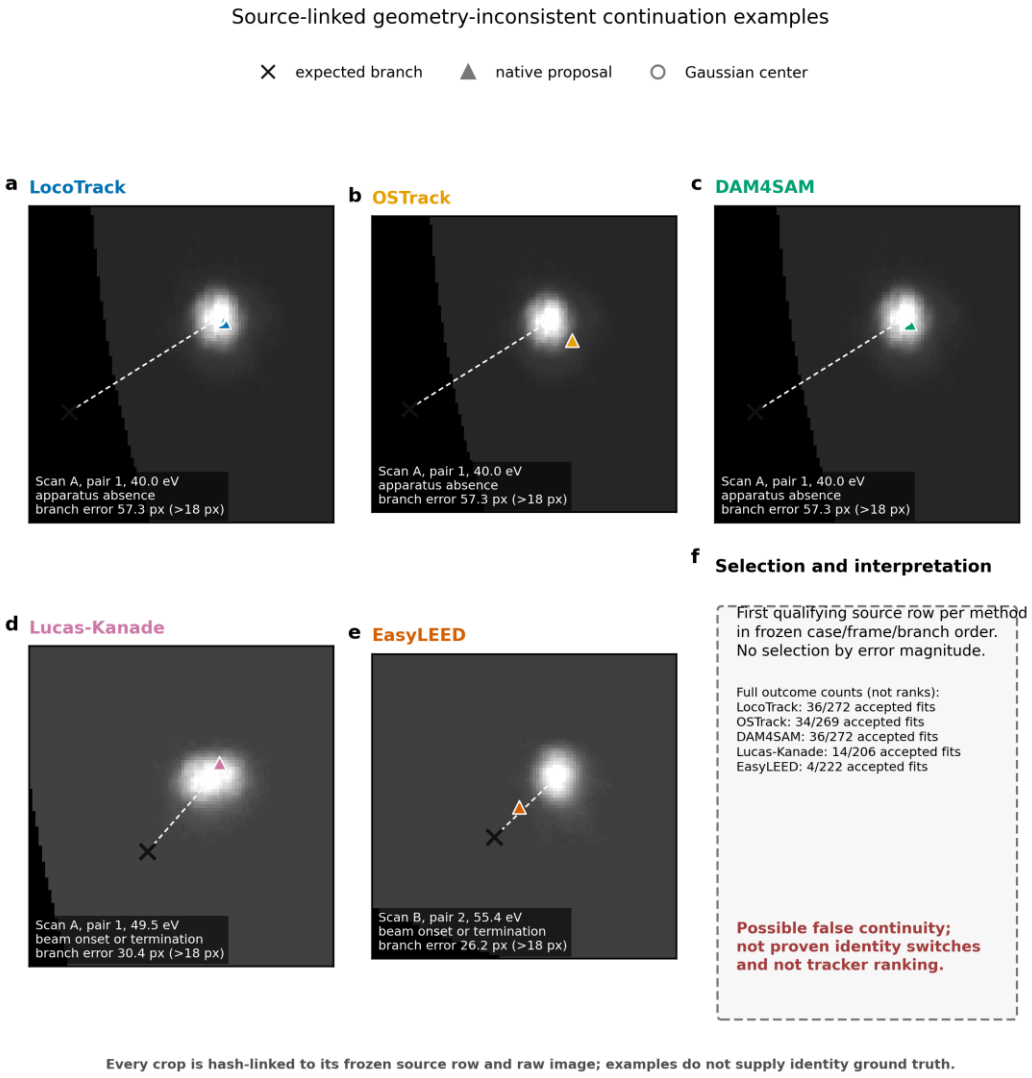


Figure S5. a–e, LocoTrack, OSTrack, DAM4SAM, Lucas–Kanade and EasyLEED examples, respectively. The black cross is the frozen expected branch coordinate, the colored triangle is the native tracking proposal, and the white circle is the accepted common Gaussian center. **f,** Selection rule and the full method-specific geometry-inconsistent/accepted-fit counts. The examples are operational evidence of possible false continuity. They are not externally proven identity switches, do not constitute a ranking and are not selected to maximize a method's branch error.

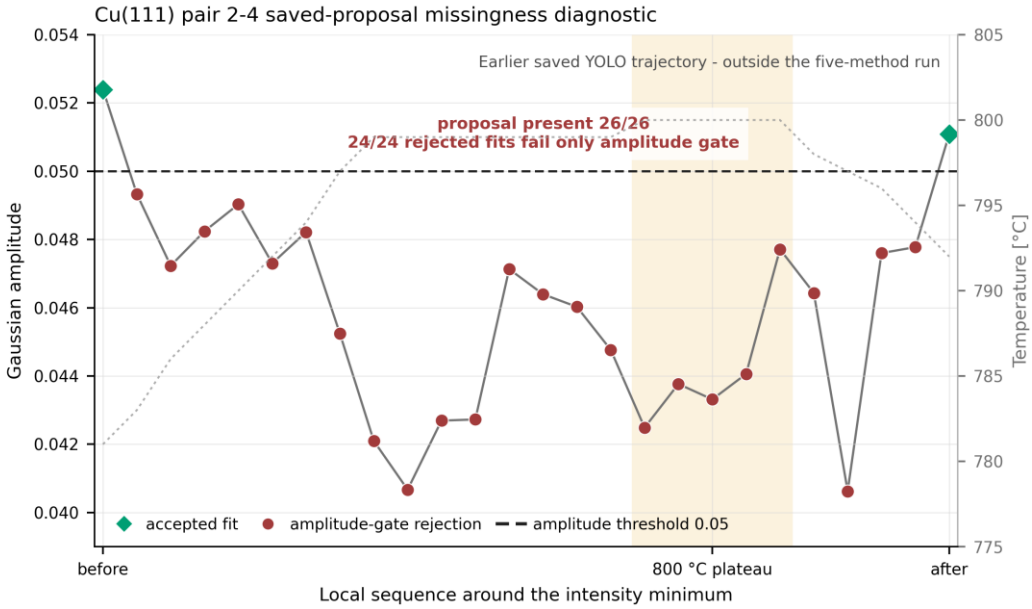


Figure S6. A saved Cu(111) pair 2-4 proposal can remain available while the common quantitative position measurement is rejected. Gaussian amplitude and the fixed acceptance threshold are shown across the 26-image context around the 800 °C plateau; accepted endpoints bracket 24 consecutive rejected fits. All rejected candidates pass the R^2, center, and width gates and fail only the uncalibrated amplitude gate. The grey dotted curve gives temperature context. This diagnostic uses an earlier saved YOLO proposal trajectory outside the frozen five-method run. It supports only the distinction between proposal availability and quantitative-measurement availability; it does not show identity loss, compare the five methods on spots 2 and 4, or define a physical LEED I(V) observable.